

FABIAN ALVAREZ-PRIMO, PhD

Regenerative Medicine · Biomaterials & Cell Therapy · Translational Research

Savannah, GA · +1 (915) 603-2117 · fpalvarez23@gmail.com

RESEARCH & TEACHING SUMMARY

Materials scientist and translational researcher whose work spans biomaterials for cardiovascular tissue engineering, non-integrating iPSC reprogramming for neuropsychiatric disease modeling, and computational structural biology. PhD in Biomedical Materials Science & Engineering (UTEP, 2020) with a metallurgy-rooted coursework core, followed by a postdoctoral fellowship applying reprogramming and transcriptomic analysis to psychiatric tissue biology. Eight peer-reviewed publications spanning biomaterials, tissue engineering, and neuropsychiatry. Subsequent industry leadership experience in large-scale, metrics-driven, quality-regulated operations directly informs a practice-oriented approach to translating bench research into scalable, compliant biomanufacturing processes.

EDUCATION

Ph.D., Biomedical Materials Science & Engineering

2017 – 2020

The University of Texas at El Paso · GPA: 3.91 · Admitted to Doctoral Candidacy, Summer 2019

- Dissertation research: Inspired Materials & Stem Cell Tissue Engineering Lab (IMSTEL)
- Graduate Certificate, 3D Engineering and Additive Manufacturing

B.S., Metallurgical & Materials Engineering

2010 – 2015

The University of Texas at El Paso

- Co-authored Appendix E: Material Properties, Wohlers Report 2016 — the additive manufacturing industry's standard annual reference report.

RESEARCH EXPERIENCE

Postdoctoral Research Fellow

2022 – 2023

Texas Tech University Health Sciences Center (TTUHSC) · Southwest Brain Bank, El Paso, TX

- Directed non-integrating episomal reprogramming of human PBMCs into induced pluripotent stem cells (iPSCs) for neuropsychiatric disease modeling, under cGMP-compliant protocols.
- Applied statistical modeling (R, Python) to transcriptomic and biomarker datasets to characterize depression-associated expression signatures across reprogrammed and postmortem neural tissue.
- Oversaw biobanking operations — sample intake, processing, storage, and chain-of-custody documentation — maintaining regulatory integrity across a multi-study pipeline.
- Collaborated across neuroscience, clinical, and molecular biology teams to coordinate study timelines, protocols, and data deliverables.
- First author on an independent transcriptomics review manuscript following change in publication direction under original PI.

Doctoral Research Assistant

2017 – 2020

Inspired Materials & Stem Cell Tissue Engineering Lab (IMSTEL), UTEP, El Paso, TX

- Led multi-year R&D project designing electrically conductive cardiovascular tissue composites for cell therapy applications, in collaboration with Rice University.
- Mentored and trained 30+ undergraduate students in electron microscopy and 3D bioprinting techniques; developed structured onboarding curricula and hands-on training protocols.
- Managed cross-functional lab operations spanning 5+ concurrent projects — equipment scheduling, reagent procurement, SOPs, and data management.
- Achieved a 30% improvement in 3D printer manufacturing precision through hardware modification; delivered funded materials projects for the US Air Force and America Makes.

Independent Research — Computational Structural Biology

2021 – Present

Self-directed

- Conducted in-silico mutagenesis and molecular docking (AutoDock Vina) study of the SERT S438T mutation, published independently with full analysis pipeline on GitHub.
- Co-author, Vargas-Medrano et al., *Frontiers in Psychiatry* (2023).

APPLIED LABORATORY & CLINICAL EXPERIENCE

Clinical Laboratory Scientist & Supervisor

2021 – 2022

West Texas Pain Institute, El Paso, TX

- Served as CLIA Laboratory Director, overseeing testing operations processing 1,300+ patient samples monthly and supervising a team of 9 laboratory professionals.
- Ensured full CLIA/COLA regulatory compliance, including DHHS guideline adherence and Drug of Abuse protocols.
- Operated and maintained LC-MS confirmatory systems, enzyme immunoassay platforms (CEDIA, DRI, QMS, EIA), and DNA next-generation sequencing instrumentation, including hands-on repair.
- Managed the laboratory information system (Stratus DX) and patient EMR records (eClinicalWorks).

APPLIED OPERATIONS LEADERSHIP

Operations Manager II, Amazon

2023 – Present

Savannah, GA

Included here as evidence of large-scale, metrics-driven, quality-regulated operational leadership directly transferable to biomanufacturing scale-up and translational program management.

- Manage a 40+ person operations team maintaining 99.7% quality compliance in a high-volume, quality-regulated environment.
- Designed and deployed a three-layer data intelligence system (browser automation, real-time analytics, staffing/rate management) replacing manual workflows across a major operational function.
- Drove an 82% improvement in throughput and reduced error/recirculation rate from 5% to under 2% through data-driven process standardization — directly analogous to yield and deviation-rate improvements sought in regulated bioprocessing.
- Built a structured training and development pipeline for 15+ team leads, established KPI systems, and led cross-functional Lean process-improvement initiatives.

Chief Scientific Officer & Co-Founder, Biosvelte

2018 – 2020

El Paso, TX

- Co-founded and scaled a biotech venture; directed product development, QC, and distribution for an evidence-based health supplement line; managed a team of 10.

PUBLICATIONS & RESEARCH OUTPUT

- 8 peer-reviewed publications spanning biomaterials, tissue engineering, and neuropsychiatry (list available upon request).
- Vargas-Medrano, J. et al. *Frontiers in Psychiatry* (2023) — co-author.
- Independent review article: cell-type-resolved transcriptomics in mood and psychotic disorders.
- Independent computational study: SERT S438T mutation, in-silico mutagenesis and molecular docking (AutoDock Vina), GitHub-published analysis pipeline.
- Alvarez-Primo, F., English, J., & Mireles, J. "Appendix E: Material Properties." *Wohlers Report 2016* (contributions from R. Wicker, D. Roberson, D. Espalin) — comparative physical and mechanical properties tables for additive manufacturing materials.

TEACHING & MENTORING

- Mentored and trained 30+ undergraduate researchers in electron microscopy and 3D bioprinting (UTEP, IMSTEL); designed onboarding curricula and hands-on training protocols.
- Built and led a 15-person structured training and development pipeline (Amazon).

HONORS & PROFESSIONAL SOCIETIES

- Tau Beta Pi Engineering Honor Society
- Alpha Chi National Honor Society
- Alpha Sigma Mu Materials Honor Society
- Society for Biomaterials — President, two terms
- Dean's List, UTEP College of Engineering (Fall 2013, Spring 2015)